

IN 1401 - Fundamentals of Databases

Tutorial 3 – EER Diagram

Question 01

The following narrative describes a simplified version of the organization of Olympic facilities planned for the summer Olympics. Draw an EER diagram that shows the entity types, attributes, relationships, and specializations for this application. State any assumptions you make.

The Olympic facilities are divided into sports complexes. Sports complexes are divided into one-sport and multisport types. Multisport complexes have areas of the complex designated for each sport with a location indicator (e.g., center, North East corner, South, etc.). A complex has a location, chief organizing individual, total occupied area, and so on. Each complex holds a series of events (e.g., the track stadium may hold many different races). For each event there is a planned date, duration, number of participants, number of officials, and so on. A roster of all officials will be maintained together with the list of events each official will be involved in. Different equipment is needed for the events (e.g., goal posts, poles, parallel bars) as well as another set of equipment is needed for the maintenance of each sports complex (e.g., cleaning equipment, grass cutters, etc.). The two types of facilities (one-sport and multisport) will have different types of information. For each type of facility, there are a number of amenities available (e.g. internet café, gym, pool, etc.) together with an approximate budget for each amenity.

Question 02

- Design a database to keep track of information for an art museum. Assume that the following requirements were collected:
- The museum has a collection of ART_OBJECTS. Each ART_OBJECT has a unique IdNo, an Artist (if known), a Year (when it was created, if known), a Title, and a Description. The art objects are categorized in several ways, as discussed below.

- ART_OBJECTS are categorized based on their type. There are three main types: PAINTING, SCULPTURE, and STATUE, plus another type called OTHER to accommodate objects that do not fall into one of the three main types.
- A PAINTING has a PaintType (oil, watercolor, etc.), material on which it is DrawnOn (paper, canvas, wood, etc.), and Style (modern, abstract, etc.).
- A SCULPTURE or a STATUE has a Material from which it was created (wood, stone, etc.), Height, Weight, and Style.
- An art object in the OTHER category has a Type (print, photo, etc.) and Style.
- ART_OBJECTS are also categorized as PERMANENT_COLLECTION, which are owned by the museum (these have information on the DateAcquired, whether it is OnDisplay or stored, and Cost) or BORROWED, which has information on the PROVIDERS (from which it was borrowed), DateBorrowed, and DateReturned.
- ART_OBJECTS also have information describing their country/culture using information on country/culture of Origin (Italian, Egyptian, American, Indian, etc.) and Epoch (Renaissance, Modern, Ancient, etc.).
- The museum keeps track of ARTISTS' information, if known: Name, DateBorn (if known), DateDied (if not living), CountryOfOrigin, Epoch, MainStyle, and Description. The Name is assumed to be unique.
- Different EXHIBITIONS occur, each having a Name, StartDate, and EndDate. EXHIBITIONS are related to all the art objects that were on display during the exhibition.
- Information is kept on PROVIDERS who provide (borrowed) items to the museum, including Name (unique), Type (museum, personal, etc.), Description, Address, Phone, and current ContactPerson.
- Draw an EER schema diagram for this application. Discuss any assumptions you made, and justify your EER design choices.